NOTICE OF CHANGE

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The documentation and process measures necessary to comply with this notice shall be completed by 5 April 2000.

MIL-STD-883E
NOTICE 3
5 November 1999

DEPARTMENT OF DEFENSE

TEST METHOD STANDARD
MICROCIRCUITS

TO ALL HOLDERS OF MIL-STD-883E:

1. THE FOLLOWING TEST METHODS OF MIL-STD-883E HAVE BEEN REVISED AND SUPERSEDE THE TEST METHODS LISTED:

NEW METHOD	DATE	SUPERSEDED METHOD	DATE
3023.1	5 November 1999	3023	19 August 1994

2. THE FOLLOWING PAGES OF MIL-STD-883E HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

METHOD	NEW PAGE	DATE	SUPERSEDED PAGE	DATE
---	iii	31 December 1996	iii	REPRINTED WITHOUT CHANGE
---	iv	24 August 1998	iv	REPRINTED WITHOUT CHANGE
---	v	5 November 1999	v	31 December 1996
---	vi	31 December 1996	vi	REPRINTED WITHOUT CHANGE
2009.9	5	19 August 1994	5	REPRINTED WITHOUT CHANGE
	6	5 November 1999	6	19 August 1994
2020.7	1	5 November 1999	1	22 March 1989
	2	5 November 1999	2	22 March 1989
2032.1	5	5 November 1999	3	1 June 1993
	6	5 November 1999	4	1 June 1993
	7	1 June 1993	7	REPRINTED WITHOUT CHANGE
	8	5 November 1999	8	1 June 1993
	13	5 November 1999	13	REPRINTED WITHOUT CHANGE
	13a	5 November 1999	13a	1 June 1993
	31	5 November 1999	31	1 June 1993
	32	5 November 1999	32	1 June 1993
	44a	5 November 1999	44a	1 June 1993
	57	1 June 1993	57	REPRINTED WITHOUT CHANGE
	58	5 November 1999	58	1 June 1993
5011.4	7	5 November 1999	7	31 October 1995
	8	31 October 1995	8	REPRINTED WITHOUT CHANGE

AMSC N/A FSC 5962

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3. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

4. Holders of MIL-STD-883E will verify that page changes, additions, and corrections indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the military standard is completely revised or canceled.

NOTE: The margins of this notice are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous notice were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous notice.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA-NA
DLA - CC

Preparing activity:
DLA - CC

Review activities
Army - AR, MI, SM
Navy - OS, SH, TD, AS, CG, MC
Air Force - 19, 99

(Project 5962-1855)

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2025.4	Adhesion of lead finish
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2029	Ceramic chip carrier bond strength
2030	Ultrasonic inspection of die attach
2031.1	Flip chip pull-off test
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TEST METHODS

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5001	Parameter mean value control
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5003	Failure analysis procedures for microcircuits
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5005.13	Qualification and quality conformance procedures
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5008.8	Test procedures for hybrid and multichip microcircuits
5009.1	Destructive physical analysis
5010.3	Test procedures for custom monolithic microcircuits
5011.4	Evaluation and acceptance procedures for polymeric adhesives.
5012.1	Fault coverage measurement for digital microcircuits.
5013	Wafer fabrication control and wafer acceptance procedures for processed GaAs wafers.

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c. Radial cracks that exhibit the following:

1. Cracks that do not originate at the lead (see figures 2009-5a and 2009-5b).

2. Three or more cracks that extend beyond the midpoints of distance from the lead to the case (see figure 2009-5c).

3. Two cracks that extend beyond the midpoint of the distance from the lead to the case and that lie within the same quadrant (see figure 2009-5d).

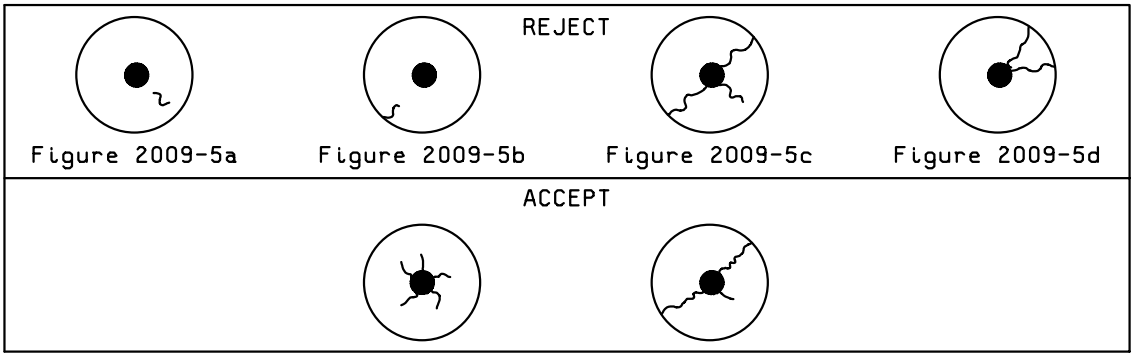


FIGURE 2009-5. Radial Cracks.

d. Any chip-out that penetrates the sealing glass deeper than the glass meniscus plane. The glass meniscus is defined as that area of glass that wicks up the lead or terminal. Exposed base metal as a result of meniscus chip outs is acceptable, provided that the exposed area is no deeper than 0.010 inch (see figure 2009-6).

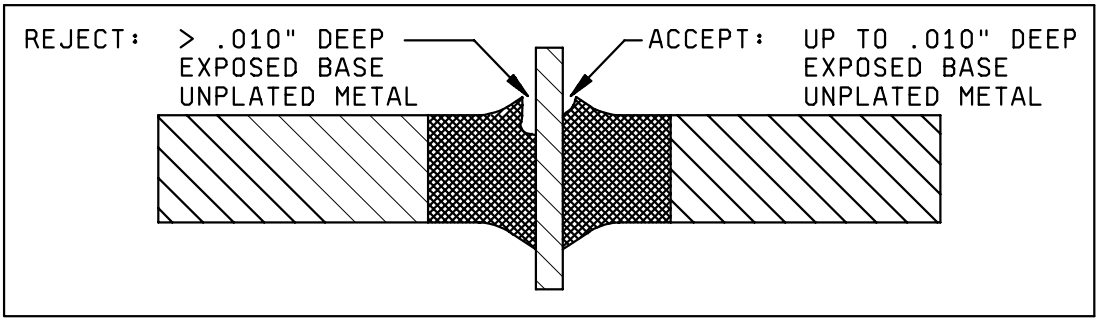


FIGURE 2009-6. Chip-outs.

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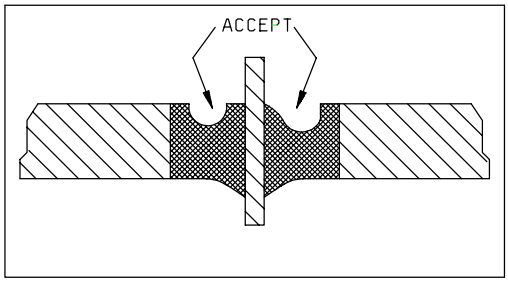


FIGURE 2009-7a. Surface bubbles.

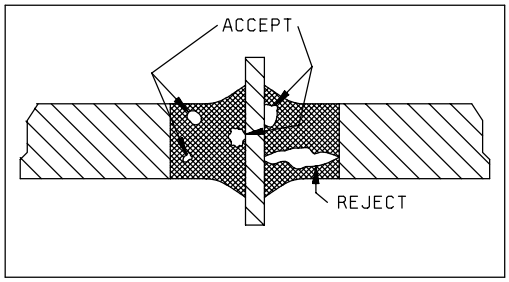


FIGURE 2009-7b. Subsurface bubbles.

e. Surface bubbles that exceed the following:

* 1. Open bubbles in the glass seal that exceed 5 mils in diameter (see Figure 2009-7a). For packages with a glass-filled header (i.e., TO-5), open bubbles that exceed 10 mils diameter, or an open bubble that exceeds 5 mils diameter situated closer than 10 mils to a lead.

2. Open bubbles in strings or clusters that exceed 2/3 of the distance between the lead and the package wall.

f. Subsurface bubbles that exceed the following:

1. Large bubbles or voids that exceed 1/3 of the glass sealing area (see Figure 2009-8a).

2. Single bubble or void that is larger than 2/3 of the distance between the lead and the package wall at the site of inclusion (see Figure 2009-7b and 2009-8b).

3. Two bubbles in a line totaling more than 2/3 distance from pin to case (see Figure 2009-8c).

4. Interconnecting bubbles greater than 2/3 the distance between pin and case (see Figure 2009-8d).

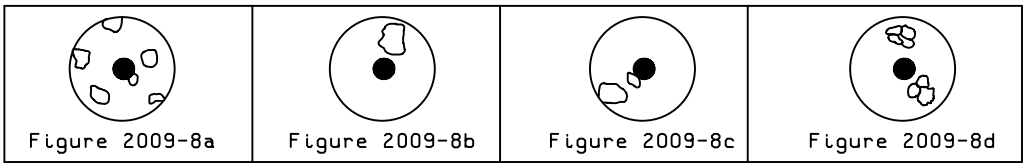


FIGURE 2009-8. Subsurface bubbles.

g. Reentrant seals that exhibit non-uniform wicking (i.e., negative meniscus) at the lead and/or body interface (see

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METHOD 2020.7

PARTICLE IMPACT NOISE DETECTION TEST

1. PURPOSE. The purpose of this test is to detect loose particles inside a device cavity. The test provides a nondestructive means of identifying those devices containing particles of sufficient mass that, upon impact with the case, excite the transducer.

2. APPARATUS. The equipment required for the particle impact noise detection (PIND) test shall consist of the following (or equivalent):

a. A threshold detector to detect particle noise voltage exceeding a preset threshold of the absolute value of 20 ±1 millivolt peak reference to system ground.

b. A vibration shaker and driver assembly capable of providing essentially sinusoidal motion to the device under test (DUT) at:

(1) Condition A: 20 g peak at 40 to 250 Hz.

(2) Condition B: 10 g peak at 60 Hz minimum.

c. PIND transducer, calibrated to a peak sensitivity of -77.5 ±3 dB in regards to one volt per microbar at a point within the frequency of 150 to 160 kHz.

d. A sensitivity test unit (STU) (see figure 2020-1) for periodic assessment of the PIND system performance. The STU shall consist of a transducer with the same tolerances as the PIND transducer and a circuit to excite the transducer with a 250 microvolt ±20 percent pulse. The STU shall produce a pulse of about 20 mV peak on the oscilloscope when the transducer is coupled to the PIND transducer with attachment medium.

e. PIND electronics, consisting of an amplifier with a gain of 60 ±2 dB centered at the frequency of peak sensitivity of the PIND transducer. The noise at the output of the amplifier shall not exceed 10 mV peak.

f. Attachment medium. The attachment medium used to attach the DUT to the PIND transducer shall be the same attachment medium as used for the STU test.

g. Shock mechanism or tool capable of imparting shock pulses of 1,000 ±200 g peak to the DUT. The duration of the main shock shall not exceed 100 µs. If an integral co-test shock system is used the shaker vibration may be interrupted or perturbed for period of time not to exceed 250 ms from initiation of the last shock pulse in the sequence. The co-test duration shall be measured at the 50 ±5 percent point.

3. PROCEDURES.

* 3.1 Test equipment setup. Shaker drive frequency and amplitude shall be adjusted to the specified conditions based on cavity size of the DUT (for condition A, see table I herein). The shock pulse shall be adjusted to provide 1,000 ±200 g peak to the DUT.

3.2 Test equipment checkout. The test equipment checkout shall be performed a minimum of one time per operation shift. Failure of the system to meet checkout requirements shall require retest of all devices tested subsequent to the last successful system checkout.

3.2.1 Shaker drive system checkout. The drive system shall achieve the shaker frequency and the shaker amplitude specified. The drive system shall be calibrated so that the frequency settings are within ±8 percent and the amplitude vibration setting are within ±10 percent of the nominal values. If a visual displacement monitor is affixed to the transducer, it may be used for amplitudes between 0.04 and 0.12 inch (1.02 and 3.05 mm). An accelerometer may be used over the entire range of amplitudes and shall be used below amplitudes of 0.040 inch (1.02 mm).

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3.2.2 Detection system checkout. With the shaker deenergized, the STU transducer shall be mounted face-to-face and coaxial with the PIND transducer using the attachment medium used for testing the devices. The STU shall be activated several times to verify low level signal pulse visual and threshold detection on the oscilloscope. Not every application of the STU will produce the required amplitude. All pulses which are greater than 20 mV shall activate the detector.

3.2.3 System noise verification. System noise will appear as a fairly constant band and must not exceed 20 millivolts peak to peak when observed for a period of 30 to 60 seconds.

3.3 Test sequence. The following sequence of operations (a through i) constitute one test cycle or run.

a. 3 pre-test shocks.

b. Vibration 3 ±1 seconds.

c. 3 co-test shocks.

d. Vibration 3 ±1 seconds.

e. 3 co-test shocks.

f. Vibration 3 ±1 seconds.

g. 3 co-test shocks.

h. Vibration 3 ±1 seconds.

i. Accept or reject.

* 3.3.1 Mounting requirements. Special precautions (e.g., in mounting, grounding of DUT leads, or grounding of test operator) shall be taken as necessary to prevent electrostatic damage to the DUT. Batch testing is prohibited.

Most part types will mount directly to the transducer via the attachment medium. Parts shall be mounted with the largest flat surface against the transducer at the center or axis of the transducer for maximum sensitivity. Where more than one large surface exists, the one that is the thinnest in section or has the most uniform thickness shall be mounted toward the transducer, e.g., flat packs are mounted top down against the transducer. Small axial-lead, right circular cylindrical parts are mounted with their axis horizontal and the side of the cylinder against the transducer. Parts with unusual shapes may require special fixtures. Such fixtures shall have the following properties:

(1) Low mass.

(2) High acoustic transmission (aluminum alloy 7075 works well).

(3) Full transducer surface contact, especially at the center.

(4) Maximum practical surface contact with test part.

(5) No moving parts.

(6) Suitable for attachment medium mounting.

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(34) Resistor ladder rung is that portion of a resistor ladder structure intended to be laser trimmed to result in an incremental change in resistance.

(35) Resistor loop is a resistor structure resembling a loop in appearance that can be trimmed. A coarse loop structure is one in which trimming results in a large resistance change (one that can cause an out-of-tolerance condition to occur). A fine loop structure is one in which trimming results in a small resistance change (one that cannot cause an out-of-tolerance condition to occur).

(36) Resistor material, self passivating is one on which a conformal insulating layer can be thermally grown (such as tantalum nitride on which tantalum pentoxide is grown).

(37) Scorching is discoloration of laser trimmed thin film resistor material without alteration of its physical form.

(38) Scratch, metallization is any tearing defect, including probe marks, in the surface of the metallization. A mar on the metallization surface is not considered to be a scratch.

(39) Scratch, resistor is any tearing defect in the resistor film. A mar on the resistor surface is not considered to be a scratch.

(40) Sidebar is that portion of a resistor ladder structure to which rungs are attached. Sidebars are not intended to be laser trimmed.

(41) Substrate is the supporting structural material into or upon which, or both, functional circuits are formed.

(42) Surface Acoustic Wave (SAW) element is a planar element fabricated typically using thin film manufacturing techniques on various substrate materials. Size varies as a function of frequency and design features include interdigitated fingers.

(43) Terminal is a metal area used to provide an electrical access point to functional circuitry.

(44) Thick film is conductive, resistive or dielectric material screen printed onto a substrate and fired at temperature to fuse into its final form.

(45) Thin film is conductive, resistive or dielectric material, usually less than 50,000Å in thickness, that is deposited onto a substrate by vacuum evaporation, sputtering, or other means.

(46) Underlying material is any layer of material below the top-layer metallization. This includes metallization, resistor, passivation or insulating layers, or the substrate itself.

* (47) Via is an opening in the insulating material in which a vertical conductive electrical connection from one metallization layer to another is made.

(48) Vitrification is conversion into glass or a glassy substance by heat and fusion.

(49) Void, metallization is any missing metallization where the underlying material is visible (exposed). Voids typically are caused by photolithographic, screen, or mask related defects, not by scratches.

(50) Void, resistor is any missing resistor material where the underlying material is visible (exposed). Voids typically are caused by photolithographic, screen, or mask related defects, not by scratches.

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(51) Wraparound conductor is one which extends around the edge of the substrate by design.

(52) Coupling (air) bridge is a raised layer of metallization used for interconnection that is isolated from the surface of the element by an air gap or other insulating material.

(53) Pit is a depression produced in a substrate surface typically by nonuniform deposition of metallization or by nonuniform processing such as excessively powered laser trim pulses.

(54) Substrate, hard is the inorganic, rigid material into or upon which or both, functional circuits are formed. Typical materials are alumina and silicon.

(55) Blister, metallization is a hollow bump that can be flattened.

(56) Nodule, metallization is a solid bump that cannot be flattened.

* (57) Substrate plug via is a cylinder-like volume in the substrate material filled with conductive material which makes electrical connection from contact areas on the top surface to the back surface of the substrate.

3.1 Thin film element inspection. Inspection for visual defects described in this section shall be conducted on each planar thin film passive element. The "high magnification" inspection shall be within the range of 100X to 200X for both class H and class K. The "low magnification" inspection shall be within the range of 30X to 60X for both class H and class K. When inspection is performed prior to mounting, then elements utilizing ceramic or glass type substrates, without backside metallization, shall be inspected using backlighting for conditions of hair-line voiding or bridging. Patterned substrates that have geometries of 2.0 mils or greater shall be inspected at 10X to 60X magnification.

	Class H	Class K
3.1.1	Operating metallization defects "high magnification". No element shall be acceptable that exhibits: NOTE: The metallization defect criteria contained in this section apply to operating metallization only.	
3.1.1.1	Metallization scratches.	
	a. A scratch or probe mark in the metallization, excluding bonding pads, that both exposes under-lying material anywhere along its length and leaves less than 50 percent of the original metallization width undisturbed (see 2032-1h). NOTE: These criteria do not apply to capacitors (see 3.1.1.1e). NOTE: Underlying material does not have to be exposed along the full length of the scratch.	a. Same as Class H.

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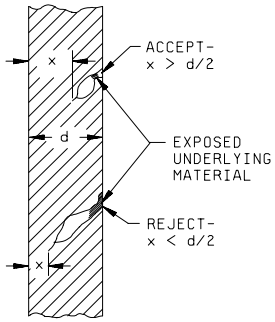


FIGURE 2032-1h. Class H metallization scratch criteria.

Class H

3.1.1.1 b. Scratch in the bonding pad area that both exposes underlying material and reduces the metallization path width, where it enters the bonding pad, and leaves less than 50 percent of its original metallization width. If two or more metallization paths enter a bonding pad, each shall be considered separately (see figure 2032-2h).

Class K

3.1.1.1 b. Less than 75 percent (see figure 2032-2k).

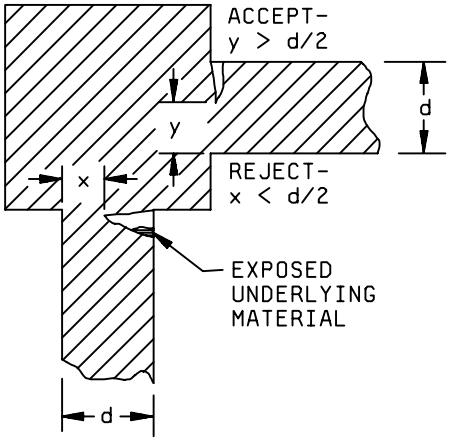


FIGURE 2032-2h. Class H metallization width reduction at bonding pad criteria.

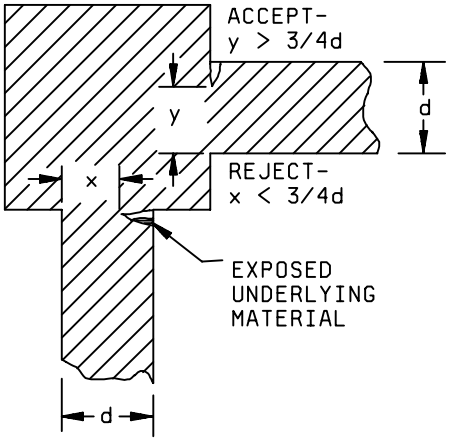


FIGURE 2032-2k. Class K metallization width reduction at bonding pad criteria.

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	Class H	Class K
* 3.1.1.1 c.	Scratch that completely crosses a metallization path and damages the surface of the surrounding passivation, glassivation, or substrate on either side.	3.1.1.1 c. Same as class H.
d.	Scratches or probe marks in the bonding pad area that expose underlying material over greater than 25 percent of the original unglassivated metallization area.	d. Same as class H.
e.	For capacitors only, a scratch in the metallization, other than in the bonding pad area, that exposes the dielectric material.	e. Same as class H.
3.1.1.2	Metallization voids.	
a.	Void(s) in the metallization, excluding bonding pads, that leaves less than 50 percent of the original metallization width undisturbed (see figure 2032-3h).	a. Same as Class H

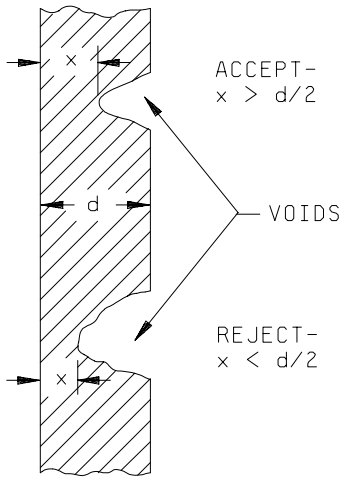


FIGURE 2032-3h. Class H metallization void criteria.

b.	Void(s) in the bonding pad area that reduces the metallization path width, where it enters the bonding pad, to less than 50 percent of its original metallization width. If two or more metallization paths enter a bonding pad, each shall be considered separately. NOTE: Figures 2032-2h and 2032-2k illustrate metallization width reduction at bonding pad criteria for scratches. Void criteria are similar.	b. Less than 75 percent

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	Class H	Class K
3.1.1.8	Metallized through-hole defects, "high magnification". No element shall be acceptable that exhibits:	
a.	Through-hole metallization that is not vertically continuous or that does not cover at least a continuous 50 percent of the inside, circumferential surface area unless by design.	a. Same as class H.
3.1.1.9	Wrap-around connection defects, "high magnification". No element shall be acceptable that exhibits:	
a.	Unmetallized area in the edges of wrap-around connections greater than 50 percent of the largest dimension of the edge metallization (see figure 2032-8h).	a. Same as class H.

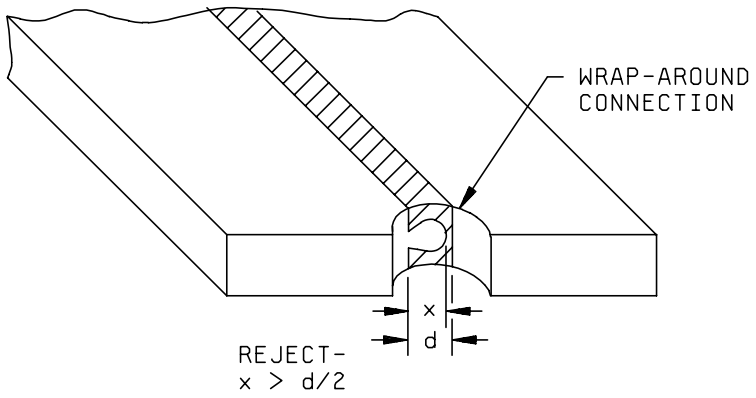


FIGURE 2032-8h. Class H wrap-around connection unmetallized area criterion.

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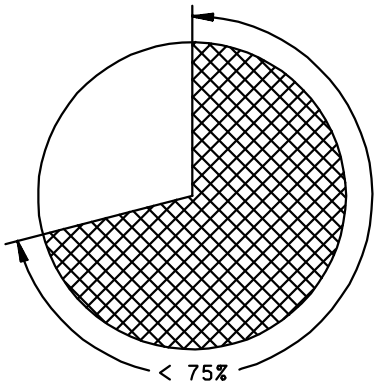
* 3.1.1.10 Substrate plug via defects, “low magnification”. When inspected from each side of the substrate, no element shall be acceptable that exhibits:

a. A complete void through the via.

b. Any lifting, peeling, or blistering of the via metallization.

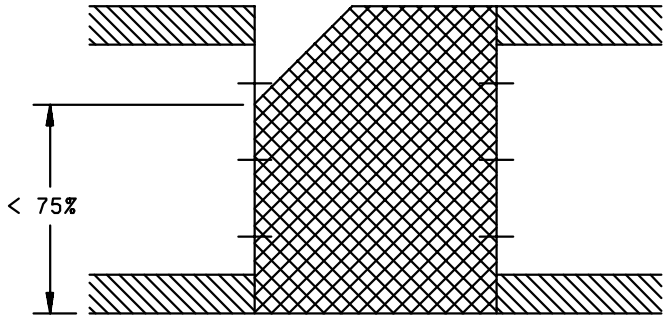
c. Via fill less than 75% of the total surface area of the via plug and less than 75% of the substrate thickness.

* NOTE: These are minimum requirements. Via flatness and other requirements shall be in accordance with the applicable detail drawings. The via fill may consist of thick film metallization.



VIA FILL < 75%

REJECT



VIA FILL < 75%

REJECT

* FIGURE 2032-8Bh. Classes H and K via plug fill criteria.

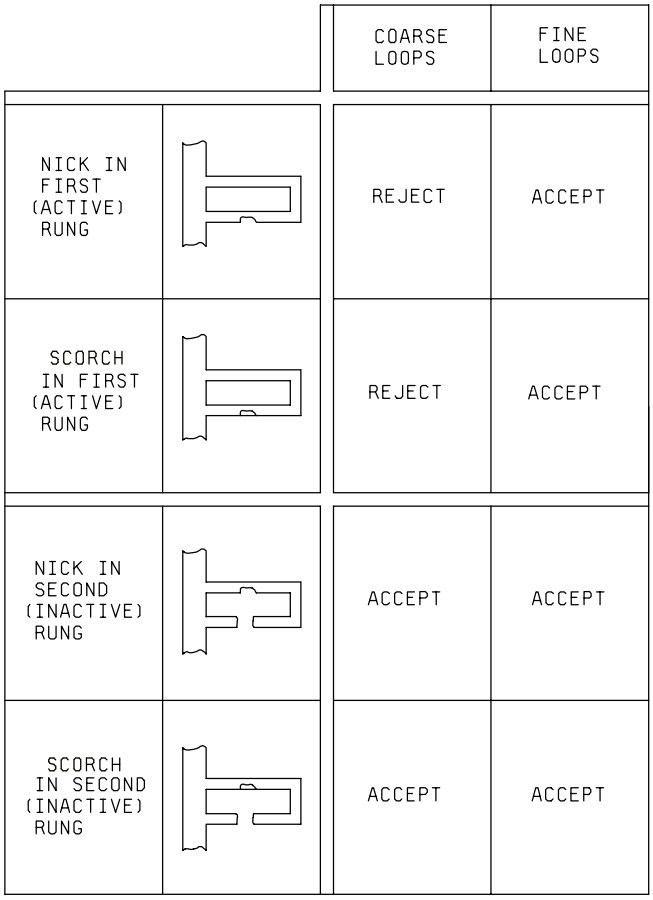
Supersedes page 13a of MIL-STD-883E

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Class H | Class K

3.1.7 d. (Continued.)
NOTE: This criteria does not apply to the second rung of a resistor loop since the second rung is inactive. This criteria does not apply to a fine loop or to a resistor structure that is comprised of fine loops (see figure 2032-28h).

NOTE: See 3.i.(35) for a definition of coarse and fine resistor loop structures. The element drawing must be referenced to determine if a given resistor loop structure is coarse or fine.

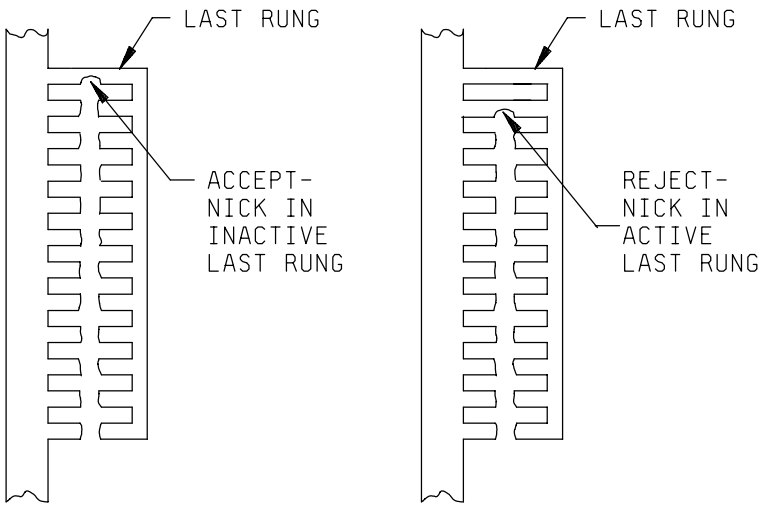


* FIGURE 2032-28h. Class H resistor loop nicking and scorching criteria exceptions.

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Class H	Class K
3.1.7 d. (Continued.) NOTE: This criterion does not apply to the last rung of a resistor ladder if the last rung is inactive (see figure 2032-29h).	



* FIGURE 2032-29h. Class H laser nicking criteria exception for the last rung of a resistor ladder.

e. A kerf or scorch which extends into a resistor ladder sidebar (see figure 2032-30h).	e. Same as class H.

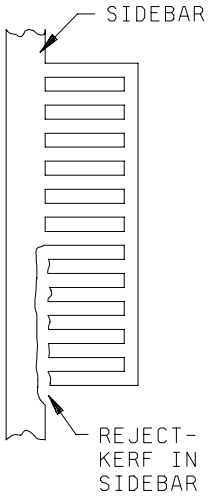


FIGURE 2032-30h. Class H resistor ladder sidebar trim criterion.

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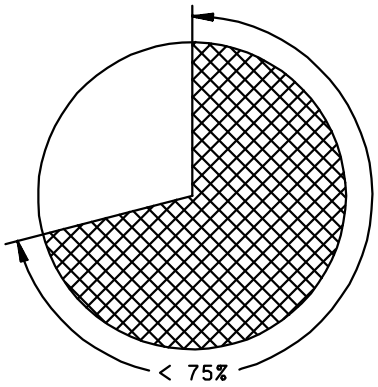
* 3.2.1.9 Substrate plug via defects, “low magnification”. When inspected from each side of the substrate, no element shall be acceptable that exhibits:

a. A complete void through the via.

b. Any lifting, peeling, or blistering of the via metallization.

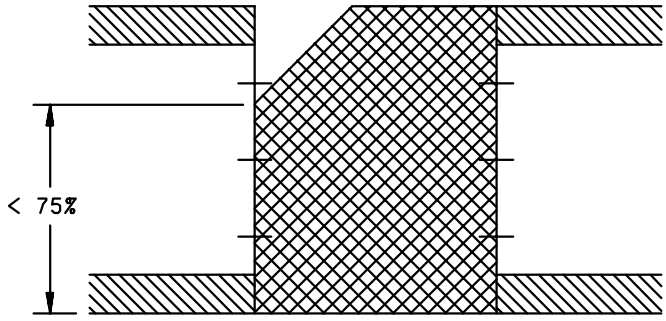
c. Via fill less than 75% of the total surface area of the via plug and less than 75% of the substrate thickness.

NOTE: These are minimum requirements. Via flatness and other requirements shall be in accordance with the applicable detail drawings.



VIA FILL < 75%

REJECT



VIA FILL < 75%

REJECT

* FIGURE 2032-43Bh. Classes H and K plug via fill criteria.

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	Class H	Class K
3.3.3 c.	Evidence of separation (delamination) of metal plates or cracks along the plane of the metal plates (see figure 2032-57h). NOTE: Narrow grooves or channel less than 1.0 mil wide that exhibit a glass-like appearance and do not expose metal plates are acceptable.	3.2.3 c. Delamination. NOTE: No delamination is allowed.

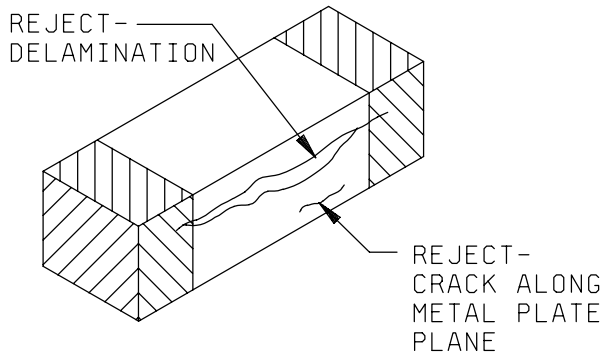


FIGURE 2032-57h. Class H delamination criteria.

d.	Crack or void in the metallization that exposes metal plates, or voids that are greater than 25 percent of the area of the metallized terminal (see figure 2032-58h).	d. Same as class H.

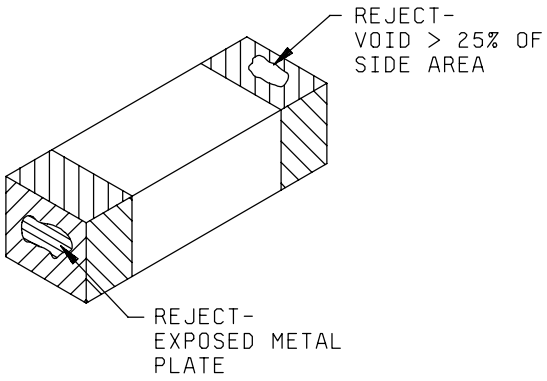


FIGURE 2032-58h. Class H termination defect criteria.

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		Class H	Class K
3.3.3	e.	Void in the metallized edges of the element that are greater than 10 percent of the metallized edge dimension, or bare corners of metallized terminals (see figure 2032-59h). NOTE: This criteria ia applicable to solder attached elements only.	3.3.3 e. Same as class H.

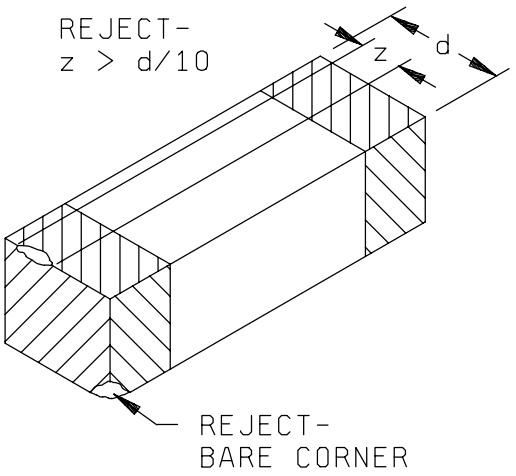


FIGURE 2032-59h. Class H metallized edge defect criteria.

	f.	Attached foreign material on the body that covers an area greater than 5.0 mils square on any side.	f. Same as class H.
3.3.4		Tantalum chip capacitor defects, "low magnification." No element shall be acceptable that exhibits:	
	a.	Flaking or peeling of the encapsulant that exposes any underlying material.	a. Same as class H.
	b.	A metallized terminal that is less than 90 percent free of encapsulant material.	b. Same as class H.
*	c.	Less than 50 percent continuous metallized terminal weld area without cracks. For capacitors with riser wires, a riser wire connection with less than 25 percent continuous weld area.	c. Same as class H.
	d.	Metallized terminal containing residue from the welding operation that is not firmly attached metallurgically to the anode cap.	d. Same as class H.

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MIL-STD-883E
NOTICE 3

METHOD 3023.1

STATIC LATCH-UP MEASUREMENTS FOR DIGITAL CMOS MICROELECTRONIC DEVICES

Latchup shall be performed in accordance with EIA/JESD78 dated March 1997. EIA/JESD78 supersedes JEDEC-STD-17.

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3.8.5.1 Thermal stability. The thermal stability of the polymeric material shall be determined by heating the specimens from room temperature to not less than 210°C, at a heating rate between 10°C/minute and 20°C/minute, in a nitrogen atmosphere with 20-30 milliliter/minute nitrogen flow. The weight loss at 200°C shall be determined.

3.8.5.2 Filler content. The filler content of polymeric materials using a filler to promote properties such as electrical or thermal conductivity shall be determined by heating the specimen from room temperature to 600°C, at a heating rate between 10°C/minute and 20°C/minute, in an air atmosphere with 20-30 milliliter/minute air flow. The temperature shall be maintained at 600°C until constant weight is obtained. It is permitted to perform 3.8.5.1, followed by heating from 210°C to 600°C as detailed above. The filler content shall be reported as weight percent of the cured specimen.

3.8.6 Outgassed materials. Ten test specimens shall be prepared using gold- or nickel-plated Kovar or ceramic packages, (dielectric materials may be prepared using aluminum coated silicon as the substrate). (The use of "leadless" packages is permitted to reduce moisture contributions due to package construction). The material shall be cured using the minimum cure schedule and shall receive the minimum pre-seal bake specified in the assembly document(s) (see 3.5.1). After a pre-seal bake, the packages shall be hermetically sealed. Only those packages that meet the fine and gross leak test requirements of test method 1014 shall be submitted for moisture content analysis. If less than 10 test specimens remain after hermetically testing, the failed packages shall be replaced by additional hermetical packages processed and tested in the same manner as the original group.

* 3.8.6.1 Testing for short term outgassing of moisture and other gaseous species. Five packages containing polymer prepared in accordance with 3.8.6 shall be heated in accordance with MIL-STD-883, method 1008, 24 hours at 150°C. The packages shall then be immediately (less than or equal to 5 minutes) inserted into the ambient gas analysis apparatus. The packages shall be subjected to ambient gas analysis in accordance with MIL-STD-883, method 1018, procedure 1. In addition to moisture, other gaseous species present in quantities greater than or equal to 100 ppmv (0.01 percent V/V) shall be reported in ppmv or percent V/V. This testing shall meet the requirements of 3.5.3.

NOTE: From the 5 packages prepared in accordance with MIL-STD-883, method 1008, only 3 packages are required to be subjected to the ambient gas analysis testing and the pass criteria of 3 packages (0 failures) shall apply (see 3.5.3). However, in the event of a failure, the testing of the remaining 2 packages shall be required in order to pass with the criteria of 5 packages (1 failure).

All polymeric materials tested shall have quantities of material equivalent in mass and exposed surface area to that of the intended application. Gold plated Kovar tabs and alumina blanks may be used as facsimile device elements. Several polymeric materials of different application may be tested in combination with each other in this test, however their combined moisture content shall not exceed 5,000 ppmv.

3.8.6.2 Testing for long term outgassing of moisture and other gaseous species. Provided that the moisture requirement of 3.5.3 has been met by packages tested in 3.8.6.1, the remaining five devices containing polymer from the group prepared in accordance with 3.8.6 shall be heated in accordance with MIL-STD-883, method 1008 for 1,000 hours at 150°C. The packages shall then be immediately (less than or equal to 5 minutes) inserted into the ambient gas analysis apparatus. The packages shall be subjected to ambient gas analysis in accordance with MIL-STD-883, method 1018, procedure 1. In addition to moisture, other gaseous species present in quantities greater than or equal to 100 ppmv (0.01 percent V/V) shall be reported in ppmv or percent V/V.

3.8.7 Ionic impurities. A water-extract analysis shall be performed to determine the level of ionic contamination in the cured polymeric material. The total ion content (specific electrical conductance) and the specific ionic content for the hydrogen (pH), chloride, sodium, fluoride and potassium ions shall be measured. Other ions present in quantities > 5 ppm shall also be reported in ppm. The methods of analysis submitted in the following paragraphs are suggested techniques. Alternate methods of analysis may be selected where it can be shown that the techniques are equivalent and the method of analysis is approved by the qualifying activity.

3.8.7.1 Sample preparation. Adequate material shall be cured to obtain 3 gram samples of polymer following grinding, for final preparation. The material shall be cured on teflon or other inert surface in a forced draft oven. When possible the cured specimen shall be removed from the curing substrate and ground to 60-100 mesh particles; polymeric film samples less than or equal to 0.025 cm thick shall be cured and cut into less than or equal to 0.25 cm^2 samples; gels or low modulus materials may be cast directly into the flat bottom of the sample flask for the extraction. Smaller sample sizes may be selected where it can be shown that the accuracy of the test method has not changed.

Supersedes page 7 of MIL-STD-883E

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3.8.7.2 Extraction procedure. 3 grams (equivalent resin) of the ground or cut equivalent polymer shall be added to a cleaned; tarred, 250-ml flasks made of pyrex, or equivalent. The weight of the cured material in each flask shall be recorded to the nearest milligram. 150.0 grams of deionized water with a measured specific conductance less than or equal to 0.1 millisiemens/meter (specific resistivity greater than or equal to 1.0 megohm-centimeter) shall be added to the flask. A blank shall be prepared by adding 150.0 grams of the deionized water and a boiling chip to a second 250-ml flask. The flasks shall be refluxed for 20 hours.

NOTE: 1.0 mho = 1.0 siemens; 1.0 mho/cm = 100.0 siemens/meter.

3.8.7.3 Measurement of ionic content.

3.8.7.3.1 Total ionic content. The total extractable ionic content shall be determined by measuring the specific electrical conductance of the water-extract samples and the blank using a conductivity meter with an immersion conductivity cell having a cell constant of 0.01/centimeter (alternatively 0.1 cm^{-1} to adjust for proper analysis of the solution). The total ionic content, in millisiemens/meter, shall be obtained by subtracting the specific conductance of the blank from the specific conductance of the samples.

3.8.7.3.2 Hydrogen ion content (pH). The pH of the water extract shall be determined using a pH meter with a standard combination electrode.

3.8.7.3.3 Specific ion analysis. Specific ion analysis of the water extract shall be conducted using ion chromatography or a demonstrated equivalent. The ion concentrations in the extract shall be converted to the sample extractable concentrations by multiplying the ratio of the deionized water weight (W) to polymer sample weight (S); that is, by (W/S). The chloride, sodium, fluoride and potassium ion levels and all other ions detected in quantities > 5 ppm shall be reported in ppm.

3.8.8 Bond strength. The bond strength of the polymeric material shall be determined in accordance with 3.8.8.1, 3.8.8.2 or 3.8.8.3 below. As a minimum, five elements shall be tested to failure at the following conditions:

a. At 25°C.

b. At 25°C after 1,000 hours at 150°C in an air or nitrogen ambient.

The average bond strength at each test condition shall be determined in kilograms (force).

3.8.8.1 Bond strength. The bond strength shall be determined in accordance with method 2019 of MIL-STD-883. A gold-metallized substrate or a gold- or nickel-plated package shall be used as the bonding surface for bond strength testing.

3.8.8.1.1 Type I materials. Suppliers shall use 0.08 inch-square (0.2 centimeter-square) gold-plated Kovar tabs.

3.8.8.1.2 Type II materials. Suppliers shall use 0.08 inch-square (0.2 centimeter-square) alumina chips.

3.8.8.2 Bond strength. The bond strength may be determined in accordance with ASTM D1002 as an alternative to test method 2019.

3.8.8.3 Molding compounds or encapsulants. Molding compounds or encapsulants shall be tested in accordance with MIL-STD-883, test method 1034.

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	MIL-STD-883E Notice 3	991105

3. DOCUMENT TITLE

Test Method Standard Microcircuits

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
Mr. Jeff Bowling	(1) Commercial (614) 692-0532	(2) AUTOVON 850-0532
c. ADDRESS *(Include Zip Code)* DSCC-VAS Post Office Box 3990 Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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